Tutorial Sheet-4

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MAL1020

Assume $\mathbb{F} = \mathbb{C}$ and $\mathcal{L}(\mathbb{F}^n)$ is the set of linear transformations from $\mathbb{F}^n$ to $\mathbb{F}^n$.

1. Find the eigenvalues and the corresponding eigenvectors of the following real matrices.

   (a) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$,

   (b) $\begin{bmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ 3 & -3 & 6 \end{bmatrix}$

2. Find the eigenvalues and the corresponding eigenvectors of the following matrices over complex numbers.

   (a) $\begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}$,

   (b) $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$,

   (c) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.

3. Find algebraic and geometric multiplicities of the eigenvalues of following matrices. Check if these matrices are diagonalizable.

   (a) $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

   (b) $B = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & -1 \\ 3 & 2 & -2 \end{bmatrix}$

4. Answers the following questions for given two matrices,

   - Find the characteristic polynomial of A.
   - Find the minimal polynomial of A.
   - Examine whether A is diagonalizable.

   (a) $\begin{bmatrix} 4 & 1 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

   (b) $\begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 2 \\ 0 & 0 & 3 \end{bmatrix}$.

5. Find inverse of matrices by using Caley-Hamilton theorem,

   (a) $\begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 5 & -4 \\ 0 & 2 & 5 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & 1 & 0 \\ -3 & 5 & 3 \\ 0 & 3 & 3 \end{bmatrix}$

6. If A is a square matrix which annihilates the polynomial $P(t) = t^2 + 5t + 1$, where annihilates stands for $P(A) = 0$. Then Is A an invertible matrix?

7. Suppose $T \in \mathcal{L}(\mathbb{C}^n)$ is invertible. Prove that $\text{Null}(T - \lambda I) = \text{Null}(T^{-1} - \lambda^{-1} I)$ for every $\lambda \in \mathbb{C}$ with $\lambda \neq 0$.

8. Consider the linear transformation $T : P_2 \to P_2$ given by $T(p(x)) = p(2x + 1)$. Prove that T is diagonalizable. Moreover Find a basis of P_2 w.r.t which T has diagonal matrix of linear transformation.

9. Suppose $S, T \in \mathcal{L}(\mathbb{C}^3)$ each have $2, 6, 7$ as eigenvalues. Prove that there exists an invertible linear transformation $P \in \mathcal{L}(\mathbb{C}^3)$ such that $S = P^{-1}TP$.

10. Find $S, T \in \mathcal{L}(\mathbb{C}^4)$ such that S, T each have $2, 6, 7$ as eigenvalues, S and T have no other eigenvalues, and there may not exist an invertible linear transformation $P \in \mathcal{L}(\mathbb{C}^4)$ such that $S = P^{-1}TP$.